Ganesh Boda

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.87
SRI CHAITANYA JR.KALASALA	Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION)	2023	938.00%
MAHARSHI VIDYALAYA -MAHABUBABAD	Class X (BOARD OF SECONDARY EDUCATION -TELANGANA STATE)	2021	10.00%

SKILLS

Languages: C, C++, Python, MySQL, CSS.

Computational Tools: Solid Edge, VHDL, Vesta, Quantum Espresso, Pymol.

Libraries: Numpy, Matplotlib, Pandas.

Developer Tools: Linux(Command Line), VS Code.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Operating Systems I & II

Scientific Computing: Computer Aided Numerical Methods I & II, Fundamentals of Scientific Computing, Computational Methods in Materials Science

Electrical: Digital Circuits, Introduction to HDL

Mathematics: Introduction to Probability, Introduction to Statistics, Linear Algebra, Transform Techniques, calculus I, calculs II, complex variables, differential equations.

PROJECTS

Numerical PDE Solver Framework | C++, Python

- Implemented finite difference solvers for advection, diffusion, heat conduction, and Burgers' equations using explicit and implicit numerical schemes.
- Developed GS-SOR, TDMA, Crank-Nicolson, ADI, MacCormack, and Lax-based solvers with stability and convergence analysis.
- Verified second-order accuracy using grid refinement, Von Neumann stability analysis, and L2 error metrics with scientific visualization.

Molecular Dynamics Simulation Engine | Python, NumPy, Computational Physics

- Built an atomistic MD simulator implementing Lennard-Jones interactions, PBC, FCC lattice generation, and Velocity Verlet numerical integration.
- Improved simulation performance using Verlet Neighbor Lists and vectorized force calculations.
- Analyzed phase behavior using RDF/MSD calculations, energy conservation, and diffusion properties.

System-Call-Aware MLFQ Scheduler & Kernel Extension in xv6

- Extended the xv6 RISC-V operating system kernel by implementing custom system calls, process tracking

mechanisms, and kernel-level statistics management.
- Designed and integrated a 4-level System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler replacing the default round-robin scheduler with priority-based execution.
- Implemented process monitoring system calls including PID tracking, parent-child process analysis, syscall accounting, scheduling level retrieval, and MLFQ statistics reporting.
- Added adaptive scheduling features such as syscall-based interactive process detection, CPU-bound process demotion, time quantum management, and global priority boosting to prevent starvation.
- Developed user-space test programs on QEMU to validate system calls, scheduler correctness, process behavior, and workload performance.

EXTRACURRICULAR

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- Active participant in Cricket and Badminton, developing teamwork, strategic thinking, coordination, and fitness.
- Participated in Inter-Department Sports Activities, contributing to a competitive and collaborative sporting environment.
- Active participant in college sports and fitness activities, maintaining a balance between academics and extracurricular engagement.